

Week 16 Capstone Presentation Rubric

CE 310 — Probability & Statistics in Civil and Architectural Engineering

Format: 4–5 slides, 4 minutes + 1 minute Q&A (5 minutes total per group)

Groups: 3 students per group; each team chose its own dataset — no two teams necessarily analyze the same topic

Total: 60 points (presentation content + delivery)

Slide Structure

Slide	Topic	Required content
1	Dataset & Client Context	Client brief summary; dataset size and source; mean predictor/outcome; a profile figure
2	Regression Findings	Model 1 vs Model 2 table; R^2 comparison; interpretation of X slope and GROUP coefficient
3	Hypothesis Test	t-test result; mean Group A vs Group B; connect t-test to regression coefficient
4	Uncertainty & Recommendation	Baseline estimate; MC P10/P90 range; sensitivity driver; one actionable insight
(5)	Optional: Deep Dive	Any additional analysis (e.g., subgroup analysis, residual plot, alternative variable)

Scoring Rubric (60 points)

Content: Slides 1–4 (40 points, 10 pts each)

Slide 1 — Dataset & Client Context (10 pts)

Criteria	Full (10)	Partial (5–7)	Minimal (1–4)	Missing (0)
Reports ANSWER_A1 (row count) and ANSWER_D1 (baseline estimate)	Yes, labeled correctly	One value present	Values wrong	Absent
Describes the dataset's profile using data (not just text)	Chart or table shown	Referenced verbally	Vague	Absent
Connects data to client context	Clear link	Attempted	Weak	Absent

Slide 2 — Regression Findings (10 pts)

Criteria	Full (10)	Partial (5–7)	Minimal (1–4)	Missing (0)
Reports Model 1 R^2 and Model 2 R^2 side by side	Both shown, labeled	One shown	Values unlabeled	Absent
Correctly states ANSWER_B1 (X slope) with units	Yes	Correct value, no units	Wrong value	Absent
Correctly interprets ANSWER_B2 (GROUP coefficient) — sign AND magnitude	Explains sign + physical meaning	Sign only	Magnitude only	Absent
If the GROUP coefficient is counterintuitive (sign or magnitude), explains why	Clear operational explanation	Mentioned without explanation	—	—

Slide 3 — Hypothesis Test (10 pts)

Criteria	Full (10)	Partial (5–7)	Minimal (1–4)	Missing (0)
Reports t-statistic and p-value	Both labeled correctly	One shown	Values present, unlabeled	Absent
States statistical conclusion in plain language (reject H_0 or fail to reject)	Correct, clearly stated	Correct, jargon-heavy	Partially correct	Absent
Connects t-test result to regression: explains why raw group mean difference $\neq$ GROUP coefficient	Clear reconciliation	Attempted	Not attempted	Absent

Slide 4 — Uncertainty & Recommendation (10 pts)

Criteria	Full (10)	Partial (5–7)	Minimal (1–4)	Missing (0)
Reports baseline estimate (ANSWER_D1) and MC P10/P90 range	Both shown	One shown	Present but wrong	Absent
Justifies the X_UNCERTAINTY assumption used in the simulation	Clearly justified	Present but vague	Unjustified	Absent
One specific, quantified, actionable recommendation	Specific + quantified	Specific, not quantified	Generic recommendation	Absent

Delivery (20 points)

Criteria	Excellent (18–20)	Good (13–17)	Adequate (7–12)	Needs Work (0–6)
Clarity — audience can follow without reading slides	Conversational, no jargon	Mostly clear	Reads slides	Unreadable
Time management — 4 min presentation $\pm$ 30 sec	Within 3.5–4.5 min	3–5 min	Outside range	—
Q&A responses — answers questions accurately	Answers from data	Reasonable attempt	Deflects	Inaccurate
Visual design — slides readable from back of room	Clean, consistent fonts/colors	Mostly readable	Cluttered	Illegible